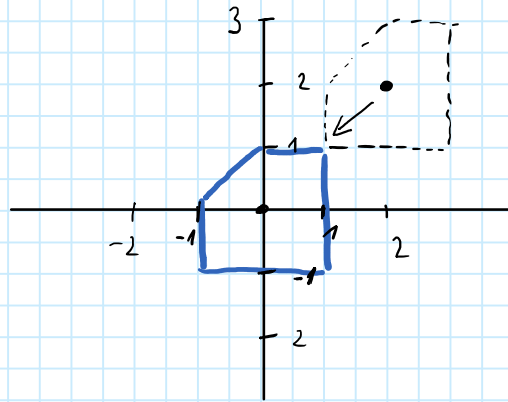


Übung Transformation - Muster

1a)

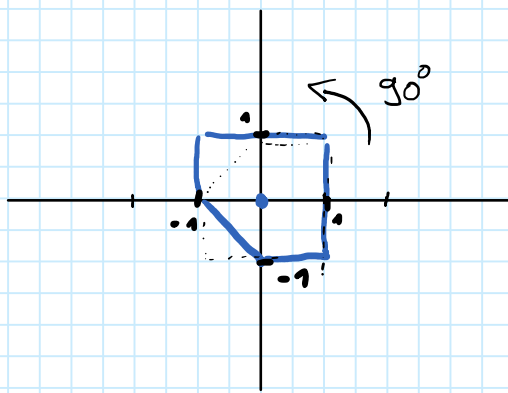
$$T = \begin{pmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{pmatrix}$$

$$T_1 = \begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{pmatrix}$$

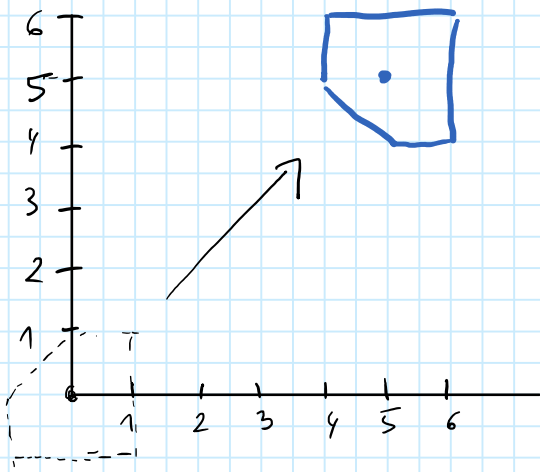


$$R = \begin{pmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$R = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$



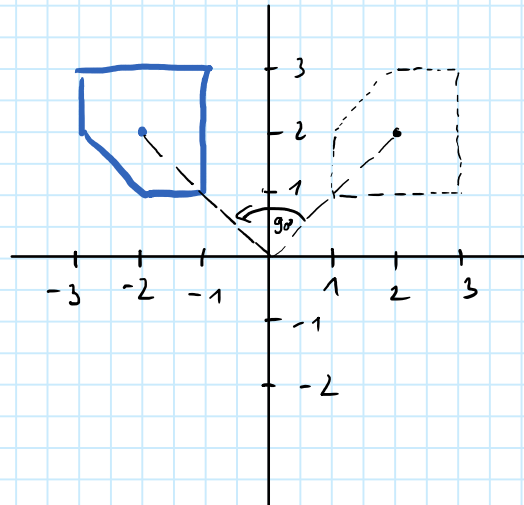
$$T_2 = \begin{pmatrix} 1 & 0 & 5 \\ 0 & 1 & 5 \\ 0 & 0 & 1 \end{pmatrix}$$



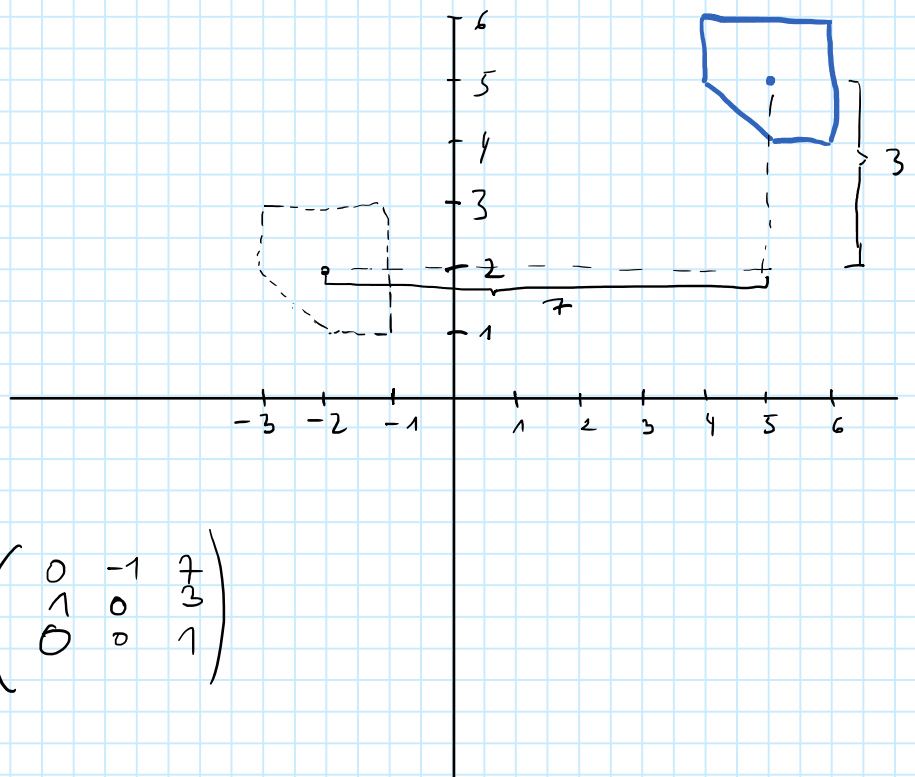
$$M = T_2 \cdot R \cdot T_1 = \begin{pmatrix} 0 & -1 & 7 \\ 1 & 0 & 3 \\ 0 & 0 & 1 \end{pmatrix}$$

1b)

$$R = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$



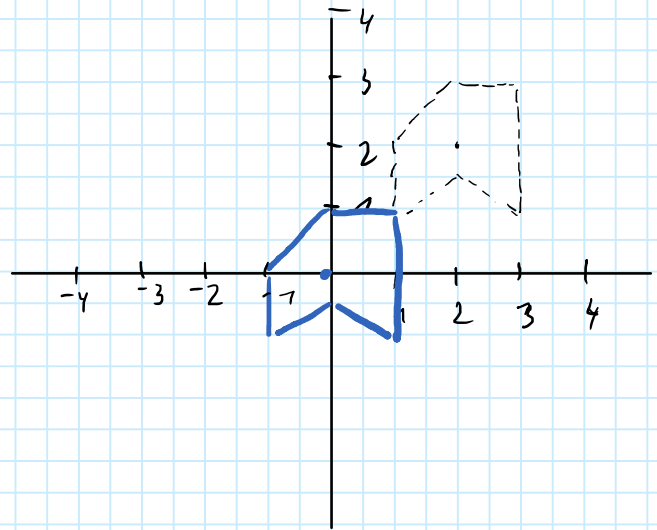
$$T = \begin{pmatrix} 1 & 0 & 7 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{pmatrix}$$



$$M = T \cdot R = \begin{pmatrix} 0 & -1 & 7 \\ 1 & 0 & 3 \\ 0 & 0 & 1 \end{pmatrix}$$

2a)

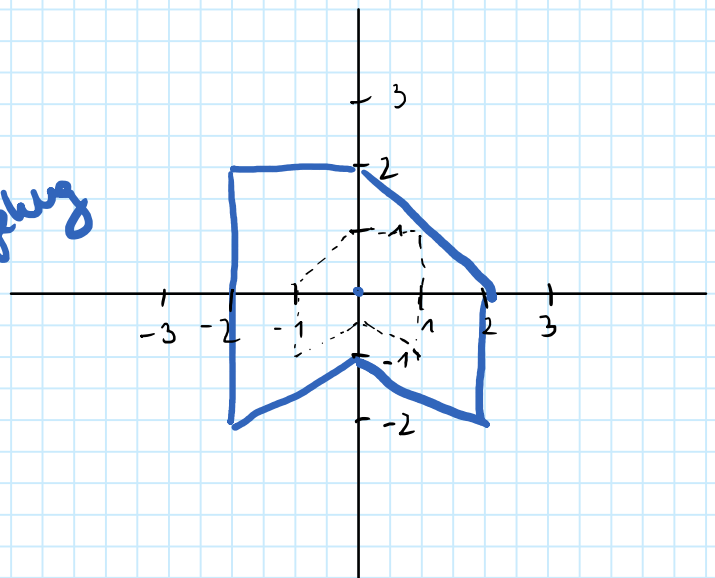
$$T_1 = \begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{pmatrix}$$



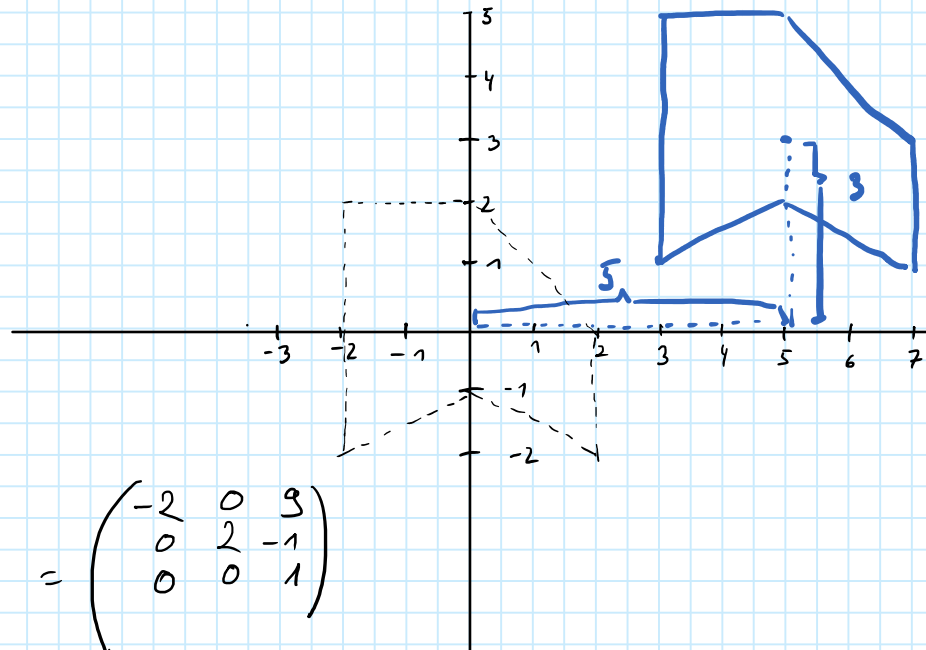
$$S = \begin{pmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$S = \begin{pmatrix} -2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

neg. Skalierung
↳ Spiegelung



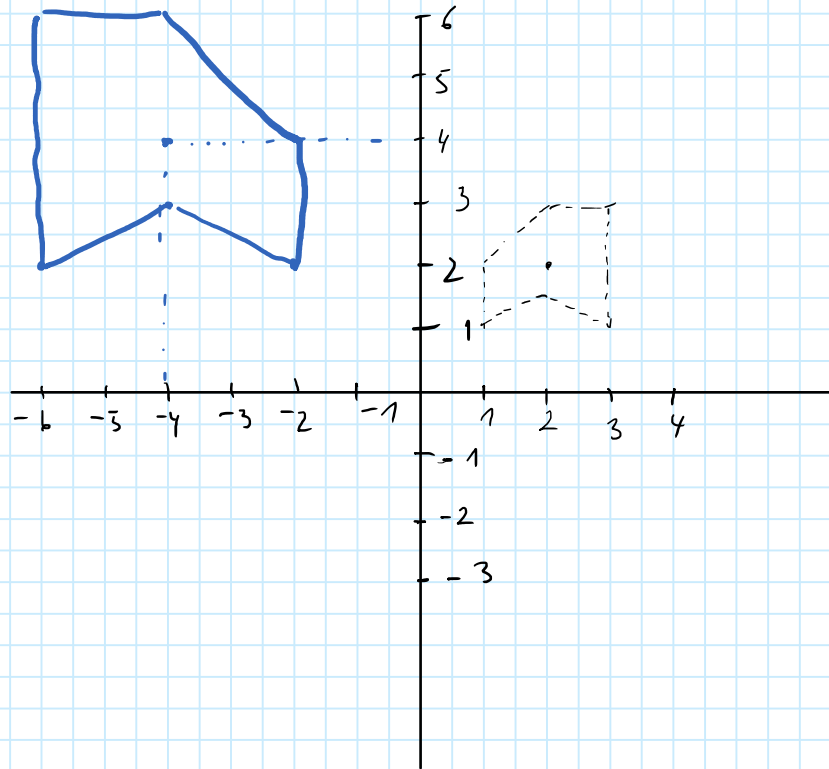
$$T_2 = \begin{pmatrix} 1 & 0 & 5 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{pmatrix}$$



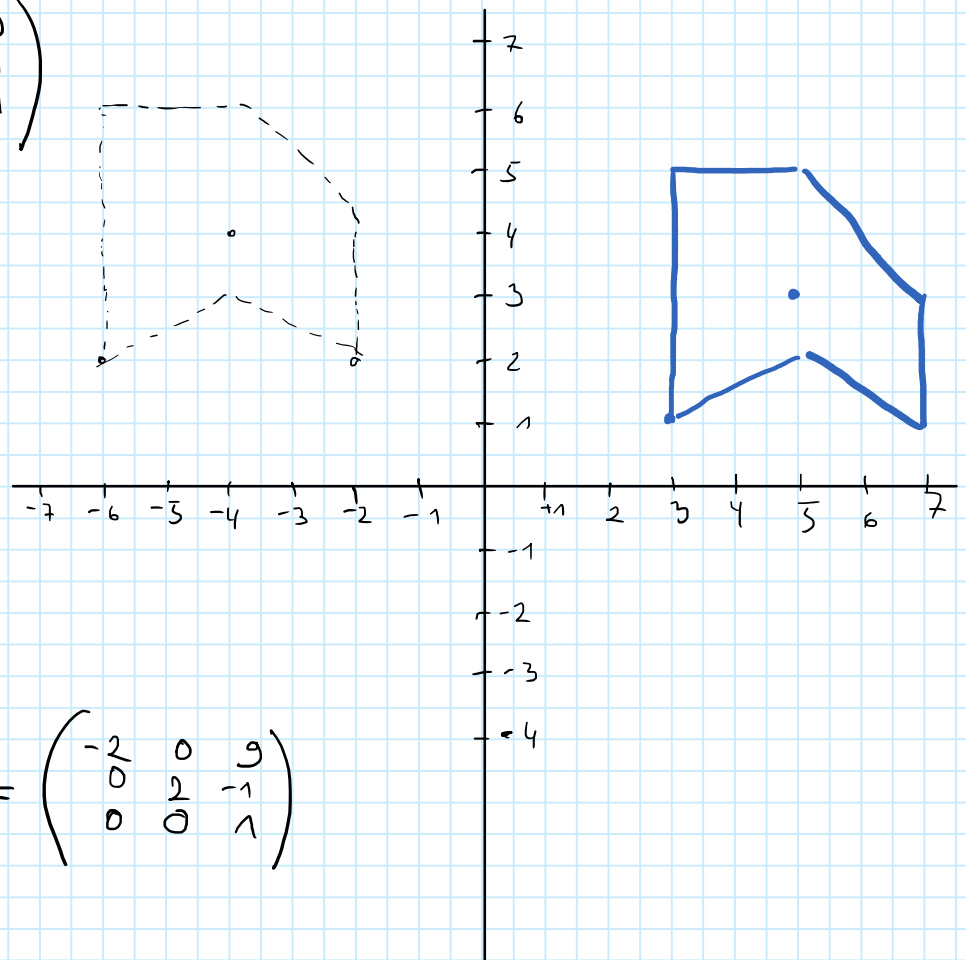
$$M = T_2 \cdot S \cdot T_1 = \begin{pmatrix} -2 & 0 & 9 \\ 0 & 2 & -1 \\ 0 & 0 & 1 \end{pmatrix}$$

2b)

$$S = \begin{pmatrix} -2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$



$$T = \begin{pmatrix} 1 & 0 & 9 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}$$



$$M = T \cdot S = \begin{pmatrix} -2 & 0 & 9 \\ 0 & 2 & -1 \\ 0 & 0 & 1 \end{pmatrix}$$

3) Möglichkeit A:

→ Ablesen der einzelnen Transformationsschritte

$$M = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & -1 \\ 0 & 0 & 1 \end{pmatrix}$$

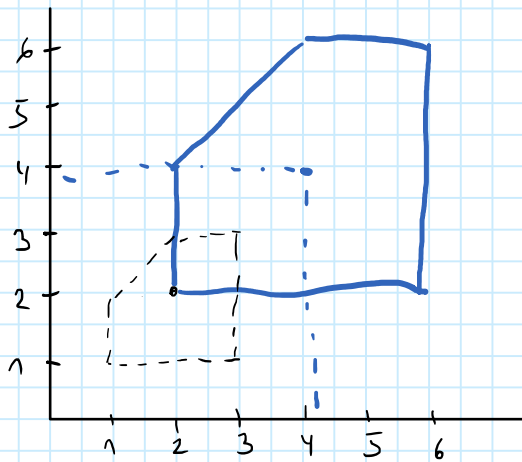
Skalierung, $s_x = s_y = 2$
Translation, $t_y = -1$

→ wir können nicht die Reihenfolge ablesen

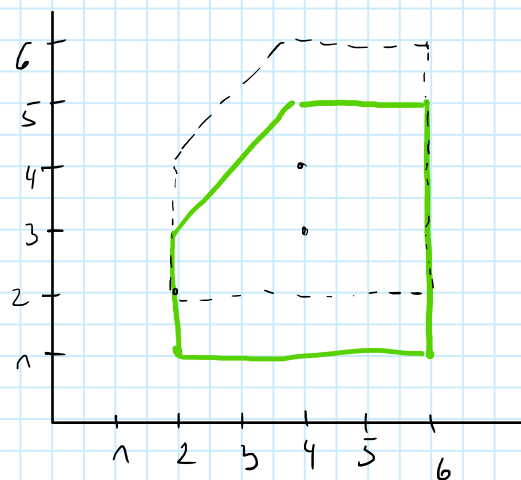
→ wir testen beide Varianten

$$M_1 = S \cdot T = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & -2 \\ 0 & 0 & 1 \end{pmatrix} \quad \text{✗}$$

$$M_2 = T \cdot S = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & -1 \\ 0 & 0 & 1 \end{pmatrix} \quad \checkmark$$



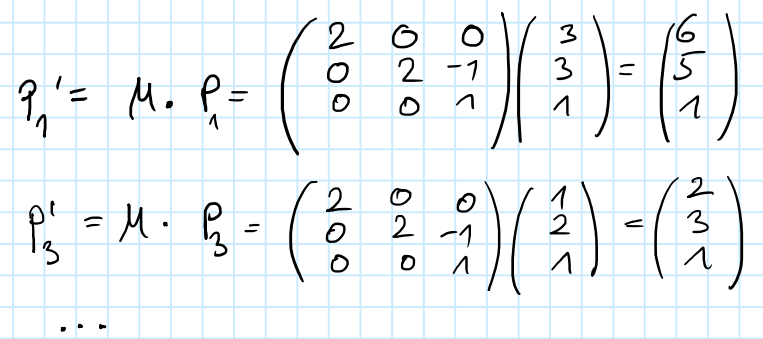
1) Skalierung



2) Translation

Möglichkeit B:

→ Exemplarisch einige Punkte im Original mit der Matrix multiplizieren



$$p'_3 = M \cdot p_3 = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & -1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}$$

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